

The Missing Layer in Public Procurement: Outcome Verification After Deployment

Abstract

Public procurement systems are designed to verify whether contracted systems are delivered according to specifications. However, they do not systematically verify whether those systems actually improve outcomes after deployment.

This paper identifies a structural gap in procurement frameworks: the absence of a defined, independent mechanism for post-deployment outcome verification. While implementation is subject to acceptance procedures and, in some cases, Independent Verification and Validation (IV&V), these mechanisms primarily operate during development and do not extend into operational outcome measurement.

As a result, public systems may be formally accepted despite failing to improve service delivery in practice. This paper argues that procurement frameworks require an additional layer—post-deployment outcome verification—to ensure that public spending translates into measurable improvements. The analysis focuses on structural design rather than specific technologies or vendors, and proposes a minimal, data-driven verification layer that can operate using existing administrative data.

1. Introduction

Public debates on government performance often focus on policy design or funding levels. However, many failures in public systems arise not from policy intent, but from delivery.

Procurement frameworks play a central role in shaping delivery outcomes. They define how systems are commissioned, implemented, and accepted. Yet a critical question remains unaddressed:

Do these systems actually improve outcomes after deployment?

In most cases, the answer is not systematically measured.

2. What Procurement Systems Verify

Standard procurement processes include:

- Contract specifications
- Milestone-based delivery
- System acceptance testing
- Compliance and audit procedures

These mechanisms ensure that:

- The system is built
- The system meets defined requirements
- The system is technically operational

In some federally funded systems, Independent Verification and Validation (IV&V), as defined under 45 CFR § 95.626, adds an additional layer of oversight during development.

However, these mechanisms share a common limitation:

They verify implementation, not outcomes.

3. The Structural Gap

The absence of post-deployment verification is not accidental—it is structural.

Across procurement frameworks:

- Vendors are responsible for system delivery
- Agencies are responsible for acceptance
- Oversight bodies conduct retrospective audits

No actor is explicitly responsible for verifying whether:

- Processing times improved
- Access increased
- Completion rates rose
- Services actually reached intended populations

Outcome verification is not assigned, and therefore not performed.

4. Consequences of the Missing Layer

This structural gap produces three systemic risks:

4.1 Formal Success, Practical Failure

Systems may pass acceptance criteria while failing to improve real-world delivery.

4.2 Lack of Accountability

Without measurable outcomes, responsibility cannot be clearly attributed.

4.3 Inefficient Public Spending

Resources are allocated and systems deployed without evidence of effectiveness.

The result is a recurring pattern: systems are implemented, but delivery outcomes remain uncertain.

5. IV&V and Its Limitations

IV&V frameworks represent an important step toward independent oversight. However, their scope is typically limited to:

- Requirements validation
- System design verification
- Implementation monitoring

They are not designed to:

- Continuously measure operational performance
- Evaluate post-deployment outcomes
- Provide ongoing delivery verification

This creates a gap between **system correctness** and **system effectiveness**.

6. The Missing Layer: Outcome Verification

To address this gap, procurement frameworks require an additional layer:

Post-Deployment Outcome Verification

This layer should:

- Operate after system go-live
- Measure real-world delivery outcomes
- Use existing administrative or public data
- Remain independent from both vendors and implementing agencies

Key characteristics:

- Minimal operational burden
 - No system integration required
 - Focus on measurable indicators (e.g., processing time, throughput, access rates)
-

7. Implications for Procurement Design

Introducing outcome verification does not require replacing existing procurement structures. Instead, it requires extending them.

A minimal integration could include:

- A defined post-deployment review period (e.g., 3–6 months)
- Outcome-based performance indicators
- Independent verification requirements linked to final acceptance or evaluation

This would shift procurement from:

“Was the system delivered?”

to:

“Did the system improve outcomes?”

8. Conclusion

Public procurement systems are effective at verifying implementation. However, they lack a mechanism to verify whether implemented systems achieve their intended outcomes.

This paper identifies this absence as a structural gap rather than an operational oversight.

Procurement verifies implementation, but not whether outcomes actually improved.

Closing this gap requires the introduction of a post-deployment outcome verification layer—one that is independent, data-driven, and integrated into procurement frameworks.

Such a layer would not replace existing oversight mechanisms, but complete them.

9. Legal and Procurement Risk Implications

The absence of post-deployment outcome verification introduces a set of legal and procurement risks that are often underrecognized.

9.1 Misalignment with Public Purpose

Public procurement is fundamentally intended to serve public outcomes. When systems are accepted based solely on implementation criteria, without verification of actual performance, a structural misalignment emerges between contractual compliance and public purpose.

9.2 Exposure to Accountability Challenges

In the absence of measurable outcome verification:

- Agencies may be unable to demonstrate that public funds produced intended results
- Oversight bodies may identify deficiencies only after systemic failures occur
- Responsibility becomes diffused across vendors, agencies, and program structures

This creates a condition in which accountability exists in principle but is difficult to operationalize.

9.3 Procurement Design Risk

When procurement frameworks do not include outcome verification:

- Acceptance criteria may not reflect real-world effectiveness
- Final payments may be decoupled from actual performance
- Future procurements may replicate unverified assumptions of success

This represents a **design-level risk**, not a failure of individual actors.

9.4 Latent Legal Exposure

While not always immediately visible, the lack of outcome verification can lead to:

- Increased audit findings and post-hoc corrective actions
- Contract disputes regarding performance versus delivery
- Potential challenges related to ineffective use of public funds

These risks do not arise from non-compliance, but from **the absence of defined outcome accountability mechanisms**.

10. Conclusion

Public procurement systems are structured to verify implementation, but not whether implemented systems achieve their intended outcomes.

This paper identifies this as a structural gap with operational and legal implications.

**Procurement verifies whether systems are delivered.
It does not verify whether outcomes improved.**

As public systems grow in scale and complexity, this gap becomes increasingly consequential—not only for delivery performance, but for accountability, audit integrity, and procurement design.

Introducing a **post-deployment outcome verification layer** would address this gap with minimal structural disruption. Such a layer would:

- Provide measurable evidence of effectiveness
- Strengthen accountability without increasing operational burden
- Align procurement outcomes with public purpose

The question for procurement systems is no longer whether implementation can be verified.

It is whether public systems can continue to operate without verifying results.